

Week 4, End-to-end machine learning pipeline

CAPP 30254, Machine Learning for Public Policy

CAPP 30254
MACHINE LEARNING
FOR PUBLIC POLICY

The end-to-end machine learning pipeline

Recall that we state our machine learning goal as:

Given the training samples $(\mathbf{x}_i, y_i)$ for $i = 1, \dots, n$ where $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \{+1, -1\}$, we want to learn a model that takes new data $\mathbf{x}$ and makes a correct class prediction $\hat{y}$.

How can we make this happen? What steps do we need to take to achieve this? *what is the actual process?*

The *end-to-end machine learning pipeline* is the entire process of building a machine learning model. It encompasses everything from raw data to evaluation and deployment.

obtaining

Define your problem

The first step of the end-to-end machine learning pipeline is to define your problem. Clearly define what you're trying to achieve with machine learning.

What's the goal? What's the target of my problem? What type of problem is this? Are you trying to predict something? Are you trying to group similar items?

classification/
regression

A well-defined problem helps you:

- choose the right data to collect
- choose the right type of model

Data collection

The second step of the end-to-end machine learning pipeline is to collect the data you need to train your model. Given your clearly defined problem, consider what kind of data you need and gather it from a database, public datasets, etc.

Considerations to make are:

- What kind of data do you need? What particular values, images, or text do you need?
numerical, categorical, etc.
values, text, images
- Where can you get the data you need? Is it already available? Do you need to scrape it from the web or use an API?
individual unit of data
- How much data do you need? Complex models and poor quality data require more data than simple models and perfect data.
deep NN
linear regression

Data cleaning

Real-world data is usually messy and imperfect. Before using it, it needs to be cleaned.

Cleaning includes:

- Handling missing values *can be handled different ways*
- Fixing inconsistencies
- Converting data to the correct type ("42" to the integer 42) *str int*
- Removing duplicates, invalid, and unneeded data
- Filtering outliers *duplicate examples and features* *is this data relevant?*
Ex. yearly income, monthly income

You'll also need to process the data so that it is in a usable format ("cat", "dog" to +1, -1).

labels

Feature selection and engineering

dependent variable

independent variables

Features are the characteristics of the examples used to make predictions in machine learning. Features are the lens through which your model sees the data. Better lenses, clearer sight.

The better the features, the better your model can learn.

Feature selection is the process of choosing which features from your data set to keep and which to leave out. The goal is to find the most useful features. Don't include: irrelevant
redundant / highly correlated

Feature engineering is the process of changing features or creating new features to make them more helpful. Creating new features is sometimes called *feature extraction*.

Feature extraction

Education level: text "college senior"

one-hot college vector: 0001

one-hot encoding:

way to represent num/class as a vec. of 0/1

zero "hot" [0, 0, 0, 1]

Feature extraction is the process of transforming raw data into a useful feature for a machine learning model.

Raw data	Extracted features
Complaint submitted to the city <i>text</i>	Topic classification (e.g., housing, sanitation), sentiment score <i>text → class</i> ~ <i>emotional tone</i>
Transcript of a city council meeting <i>text</i>	Speaker count, number of public comments, presence of specific terms (e.g., budget, safety)
Image of a street corner <i>image</i>	Presence of crosswalk, number of lanes, lighting conditions
Education level	Education level encoded numerically or as one-hot vector